

Module - III

Trusses

Truss is a structural member made-up of slender members pin connected at ends & is capable of taking loads at joint, and it is also called pin-connected frame.

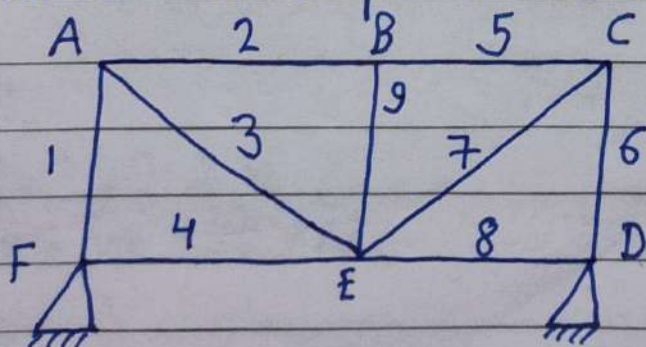
Assumptions of Truss:

1. Self weight of members is negligible.
2. The cross-section of the member is uniform.
3. The ends of the members are pin connected.
4. The load acts only at joints.

Types of Frames:

1. Perfect Truss:

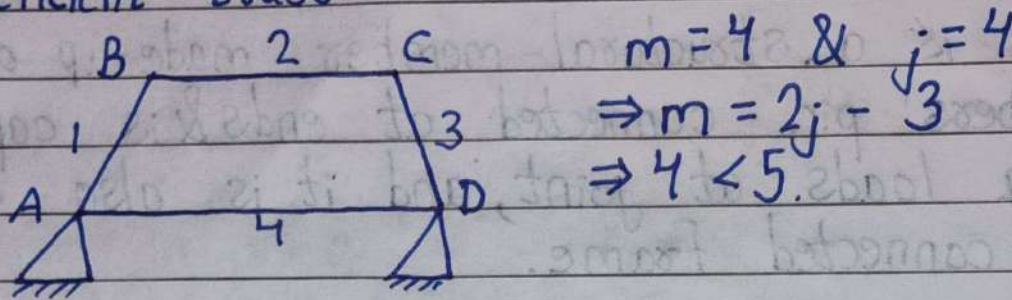
If the number of members are just sufficient to resist the load without undergoing appreciable deformation in shape is known as perfect truss.



$$\begin{aligned} m &= 9 \text{ \& } j = 6 \\ \Rightarrow m &= 2j - 3 \\ \Rightarrow 9 &= 9 \end{aligned}$$

2. Deficient Truss:

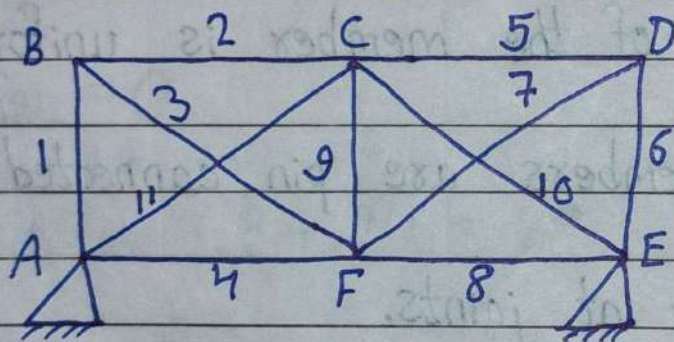
If the number of members in a truss is less than the required for perfect truss is known as deficient truss.



$$m = 4 \text{ \& } j = 4$$
$$\Rightarrow m = 2j - 3$$
$$\Rightarrow 4 < 5$$

3. Redundant Truss:

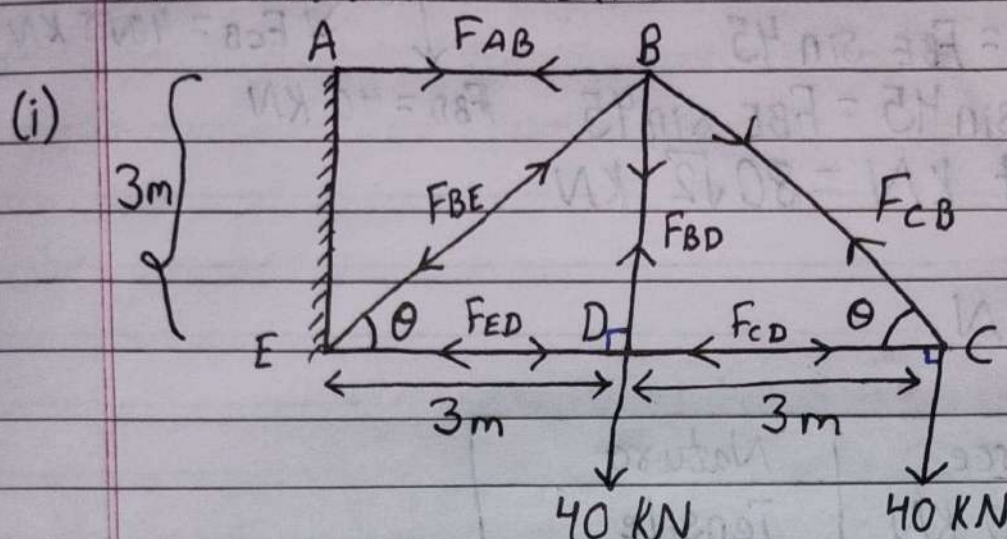
If the number of members in a truss is more than the required for perfect truss is known as redundant truss.



$$m = 11 \text{ \& } j = 6$$
$$\Rightarrow m = 2j - 3$$
$$\Rightarrow 11 > 9$$

Numexicals

Q) Find the forces in all the members of the truss and tabulate result.



Solⁿ:

$$\tan \theta = \frac{3}{3} = 1 \Rightarrow \theta = 45^\circ$$

At C,

$$\frac{F_{CB}}{\sin 90} = \frac{F_{CD}}{\sin (90+45)} = \frac{40}{\sin (90+45)}$$

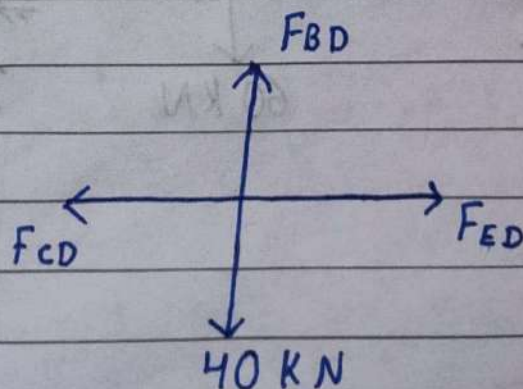
$$\Rightarrow F_{CB} = \frac{40 \sin 90}{\sin(90+45)} = 40\sqrt{2} \text{ KN}$$

$$\Rightarrow F_{CD} = \frac{40 \sin(90+45)}{\sin(90+45)} = 40 \text{ kN}$$

At D,

$$F_{BD} = 40 \text{ kN}$$

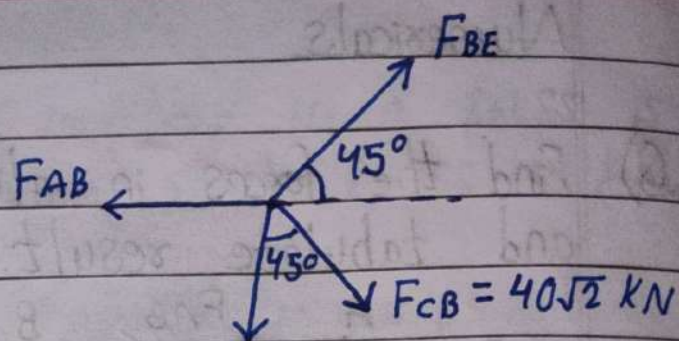
$$F_{ED} = F_{CD} = 40 \text{ kN}$$



At B,

$$F_{AB} = F_{BE} \cos 45 + F_{CB} \cos 45$$

$$\Rightarrow F_{AB} = F_{BE} \cos 45 + 40$$



$$F_{BD} + F_{CB} \sin 45 = F_{BE} \sin 45$$

$$\Rightarrow 40 + 40\sqrt{2} \cdot \sin 45 = F_{BE} \sin 45$$

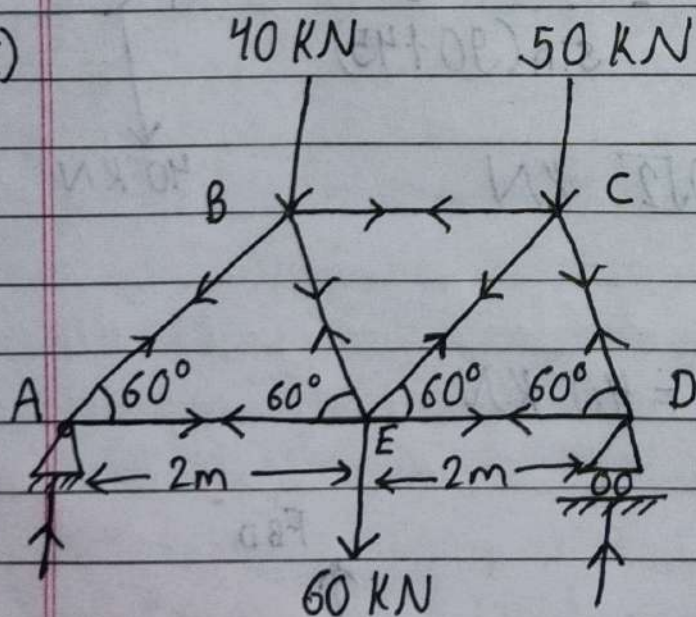
$$F_{BD} = 40 \text{ kN}$$

$$\Rightarrow F_{BE} = 113.137 \text{ kN} = 80\sqrt{2} \text{ kN}$$

$$\therefore F_{AB} = 120 \text{ kN}$$

Member	Force	Nature
AB	120 kN	Tensile
BC	$40\sqrt{2}$ kN	Tensile
CD	40 kN	Compressive
DE	40 kN	Compressive
BE	$80\sqrt{2}$ kN	Compressive
BD	40 kN	Tensile

(ii)



Solⁿ:

$$R_A + R_D = 40 + 60 + 50 = 150 \text{ kN}$$

$$\sum M_A = 0$$

$$\Rightarrow R_D \times 4 = 40 \times 2 \cos 60^\circ + 60 \times 2 + 50 \times 3$$

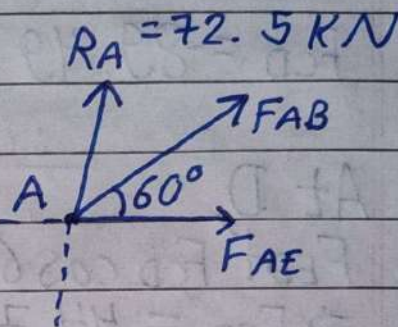
$$\Rightarrow R_D = 77.5 \text{ kN}$$

$$\therefore R_A = 72.5 \text{ kN}$$

At A,

$$R_A + F_{AB} \sin 60 = 0$$

$$\Rightarrow F_{AB} = \frac{-R_A}{\sin 60} = -83.715 \text{ kN}$$



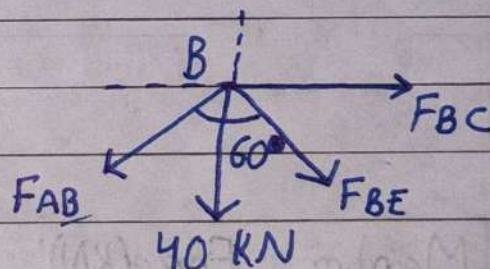
$$F_{AE} + F_{AB} \cos 60 = 0$$

$$\Rightarrow F_{AE} = 41.857 \text{ kN}$$

At B,

$$F_{AB} \cos 60 = F_{BC} + F_{BE} \cos 60$$

$$F_{BC} + F_{BE} \cos 60 = -41.857 \text{ kN}$$



$$40 + F_{AB} \sin 60 + F_{BE} \sin 60 = 0$$

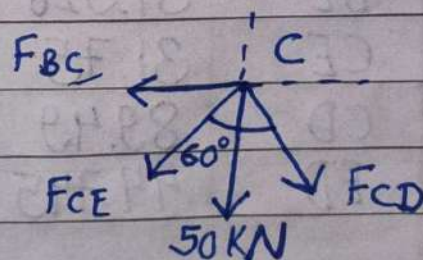
$$\Rightarrow F_{BE} = \frac{-40 - F_{AB} \sin 60}{\sin 60} = 37.526 \text{ kN}$$

$$F_{BC} = -60.62 \text{ kN}$$

At C,

$$F_{BC} + F_{CE} \cos 60 = F_{CD} \cos 60$$

$$\Rightarrow F_{CD} = \frac{-60.62 + F_{CE} \cos 60}{\cos 60}$$



$$F_{CE} \sin 60 + 50 + F_{CD} \sin 60 = 0$$

$$\Rightarrow F_{CE} \sin 60 + 50 + \left(\frac{-60.62 + F_{CE} \cos 60}{\cos 60} \right) \sin 60 = 0$$

$$\Rightarrow F_{CE} \sin 60 + 50$$

$$\Rightarrow 0.433 F_{CE} + 25 + (-60.62 + 0.5 F_{CE}) \sin 60 = 0$$

$$\Rightarrow 0.433 F_{CE} + 25 - 52.498 + 0.433 F_{CE} = 0$$

$$\Rightarrow 0.866 F_{CE} = 27.498$$

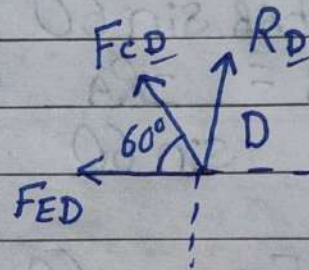
$$\Rightarrow F_{CE} = 31.75 \text{ kN}$$

$$F_{CD} = -89.49 \text{ kN}$$

At D,

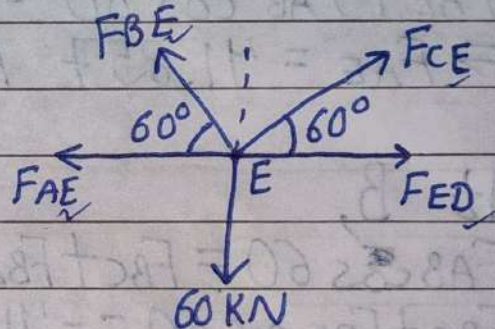
$$F_{ED} + F_{CD} \cos 60 = 0$$

$$\Rightarrow F_{ED} = 44.745 \text{ kN}$$

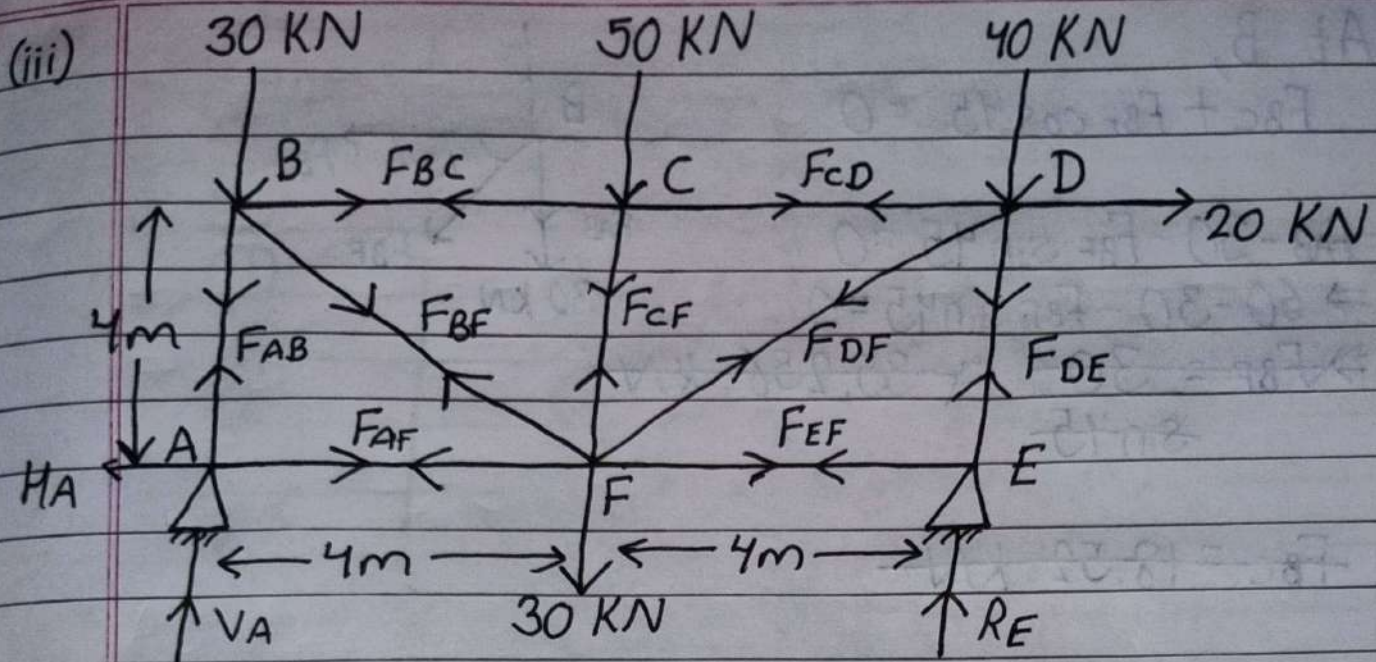


At E,

All forces found.



Member	Force(kN)	Nature
AB	83.715	Compression
AE	41.857	Tension
BC	60.62	Compression
BE	37.526	Tension
CE	31.75	Tension
CD	89.49	Compression
ED	44.745	Tension



Solⁿ:

$$H_A = 20 \text{ kN}$$

$$V_A + R_E = 30 + 50 + 40 + 30 = 150 \text{ kN}$$

$$\sum M_A = 0$$

$$\Rightarrow 30 \times 4 + 50 \times 4 + 40 \times 8 + 20 \times 4 = R_E \times 8$$

$$\Rightarrow R_E = 90 \text{ kN}$$

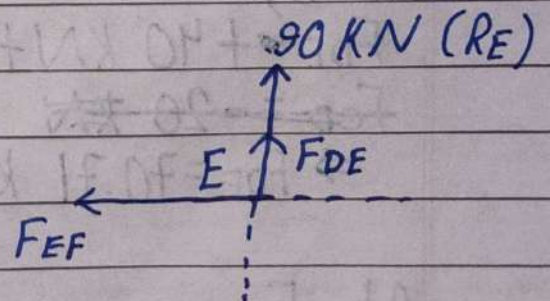
$$\therefore V_A = 150 - 90 = 60 \text{ kN}$$

At E,

$$R_E + F_{DE} = 0$$

$$\Rightarrow F_{DE} = -90 \text{ [Compressive]}$$

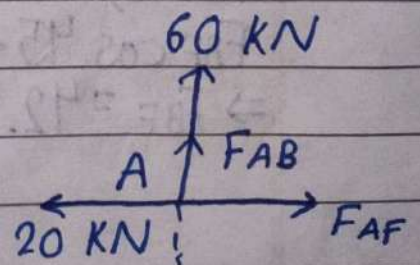
$$F_{EF} = 0$$



At A,

$$F_{AF} = H_A = 20 \text{ kN}$$

$$\& F_{AB} = -V_A = -60 \text{ kN [Compressive]}$$



At B,

$$F_{BC} + F_{BF} \cos 45 = 0$$

$$-F_{AB} - 30 - F_{BF} \sin 45 = 0$$

$$\Rightarrow 60 - 30 - F_{BF} \sin 45 = 0$$

$$\Rightarrow \cancel{F_{BF} = 30} \quad \cancel{= 35.256 \text{ kN}} \\ \sin 45$$

$$\cancel{F_{BC} = 18.52 \text{ kN}}$$

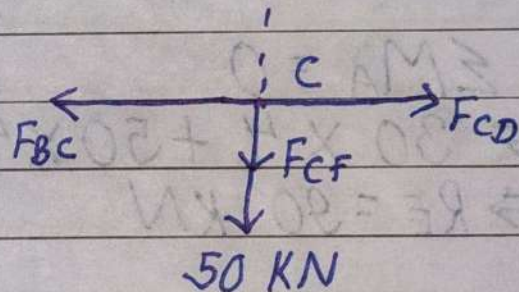
$$\Rightarrow F_{BF} = \frac{30}{\sin 45} = 42.43 \text{ kN}$$

$$F_{BC} = -30 \text{ kN [compressive]}$$

At C,

$$F_{CD} = F_{BC} = -30 \text{ kN}$$

$$F_{CF} = -50 \text{ kN}$$

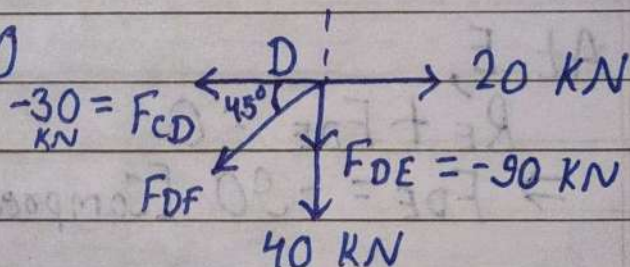


At D,

$$F_{DE} + 40 \text{ kN} + F_{DF} \sin 45 = 0$$

$$\cancel{F_{CD} = 20 \text{ kN}}$$

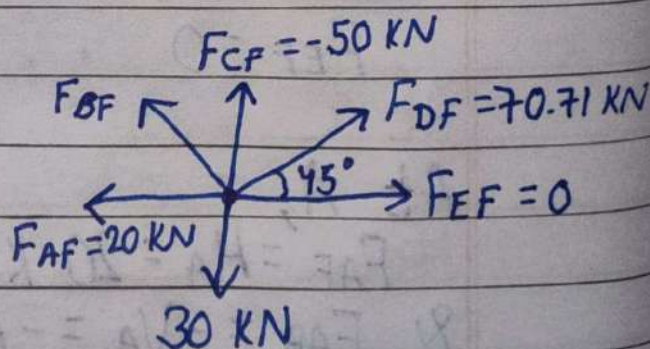
$$\Rightarrow F_{DF} = 70.71 \text{ kN}$$



At E,

$$F_{BF} \cos 45 + F_{AF} = F_{EF} + F_{DF} \cos 45$$

$$\Rightarrow F_{BF} = 42.425 \text{ kN}$$



Members	Force	Nature
AB	60 kN	Compressive
BC	30 kN	Compressive
CD	30 kN	Compressive
DE	90 kN	Compressive
EF	0 kN	Zero-force
AF	20 kN	Tensile
BF	42.425 kN	Tensile
CF	50 kN	Compressive
DF	70.71 kN	Tensile